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```

clc;
clear;
close all;
% (7,4) Hamming Code
msg = [1 0 1 1];
% Encode
code = encode(msg,7,4, 'hamming/binary');
% Introduce Single-Bit Error
rx = code;
rx(3) = ~rx(3); % Flip one bit
% Decode and Correct Error
dec = decode(rx,7,4, 'hamming/binary');
% Display Results
disp('Message Bits:')
disp(msg)
disp('Encoded Bits:')
disp(code)
disp('Received Bits (with error):')
disp(rx)
disp('Corrected Message:')
disp(dec)
% Plot
figure;
subplot(2,1,1)
stem(code, 'filled')
title('Encoded Codeword')
grid on
subplot(2,1,2)
stem(rx, 'filled')

title('Received Codeword with Single-Bit Error')
grid on

```

*Message Bits:*

1      0      1      1

*Encoded Bits:*

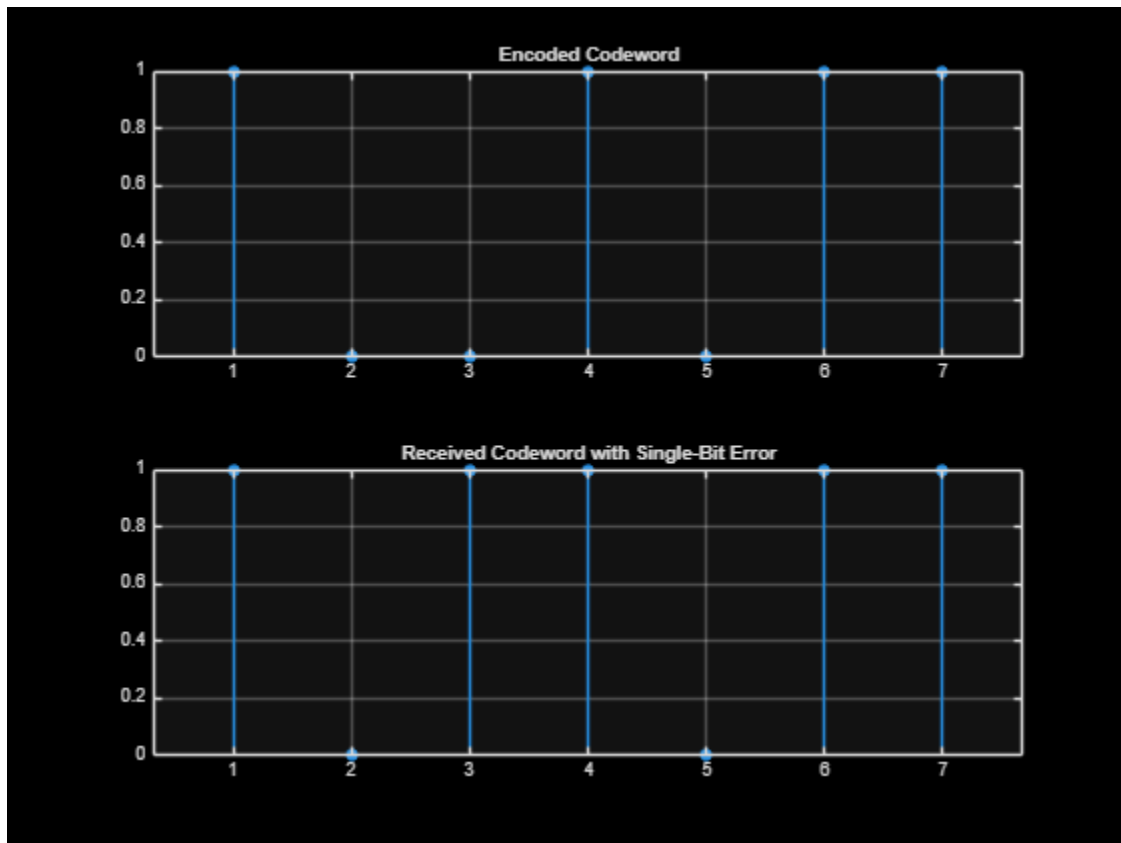
1      0      0      1      0      1      1

*Received Bits (with error):*

1      0      1      1      0      1      1

*Corrected Message:*

1      0      1      1



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